

Package ‘ScreenBEAM2’

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Type Package

Title A novel meta-analysis algorithm for functional genomics screens via Bayesian hierarchical modeling, Version 2

Version 0.1.0

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Description ScreenBEAM2 is the upgraded version of ScreenBEAM1. It can perform functional genomics screening from FASTQ file. It is a trim-free, allowing number of mismatch, BLAT-depend pipeline. Downstream analysis uses Bayesian hierarchical modeling, removed bias caused by the unbalanced number of sh/sgRNA targeting the same gene.

License Apache License

Encoding UTF-8

LazyData true

RoxygenNote 7.2.3

Depends R (>= 3.6.0),
Biobase (>= 2.38.0),
readr (>= 1.3.1),
kableExtra (>= 1.1.0),
knitr (>= 1.24),
data.table (>= 1.12.2),
graphics,
grDevices,
plyr (>= 1.8.4)

Imports MCMCglmm (>= 2.29),
RColorBrewer (>= 1.1-2),
stats (>= 3.6.0),
stringr (>= 1.4.0),
tidyr,
dplyr (>= 0.8.0),
methods,
lme4,
rmarkdown (>= 2.1),
arm (>= 1.10.1)

Suggests NetBID2

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generateEset	<i>Generate Eset by number of mismatch and total normalize number</i>
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Description

Generate Eset by number of mismatch and total normalize number

Usage

```
generateEset(m, n.mismatch = NULL, normalize = TRUE, normalize.total = 1e+06)
```

Arguments

m	matrix
n.mismatch	number of mismatch
normalize	logical, default to TRUE
normalize.total	default to 1e6

Value

an eset with

getCountByMismatch	<i>Get count matrix by number of mismatch</i>
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Description

Get count matrix by number of mismatch

Usage

```
getCountByMismatch(  
  count.dist,  
  n.mismatch,  
  normalize = TRUE,  
  normalize.total = 1e+06,  
  annotation = NULL,  
  undertermined.col = NULL,  
  ...  
)
```

Arguments

count.dist	count distribution matrix, originally outputed in count.dist.csv from Screen-BEAM.raw.count
n.mismatch	numerical,number of mismatch
normalize	logical
normalize.total	total normalize number, default to 1e6
annotation	NULL
undertermined.col	NULL
...	parameters pass to normalizeMiseqProfile

normalize.scale	<i>Normalization function</i>
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Description

Normalization function

Usage

```
normalize.scale(d, total = NULL, pseudoCount = 1)
```

Arguments

pseudoCount

normalizeMiseqProfile	<i>normalizeMiseqProfile</i>
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Description

normalizeMiseqProfile

Usage

```
normalizeMiseqProfile(
  d,
  total = NULL,
  pseudoCount = 1,
  undertermined.col = "Undetermined"
)
```

Arguments

d	data frame
total	total normalize number
pseudoCount	pseudo number to add, for eliminating zeros
undertermined.col	default to 'Undetermined'

saveCountEset	<i>saveCountEset: Save count data into different expression set with multiple number of mismatch value</i>
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Description

saveCountEset: Save count data into different expression set with multiple number of mismatch value

Usage

```
saveCountEset(m, save.path, total = 1e+06, n.mismatch = 2)
```

Arguments

m	matrix
save.path	path to save your expression sets
total	total normalization number
n.mismatch	numeric, adjust according to mapping qc report

Value

Four expression sets

ScreenBEAM.check.lib *Check sh/sgRNA library and Remove duplicated/substring sh/sgRNA*

Description

ScreenBEAM.check.lib is a pre-processing function for ScreenBEAM.raw.count. It requires the file path of the library file (.csv file, with 1st column to be RNAid, 2nd column RNA sequence, 3rd column target gene name). It will remove the duplicated/substring sh/sgRNA and store it into a list.

Usage

```
ScreenBEAM.check.lib(lib.path)
```

Arguments

lib.path	character, the file path of the library. Should be a csv file. The 1st column is RNAid, 2nd column RNA sequence and 3rd column target gene name.
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Value

Return a list object, containing whole library information after removed duplicates; The most command sh/sgRNA length; Total sh/sgRNA number; All the sh/sgRNA names; Each sh/sgRNA length; Removed sh/sgRNA information.

ScreenBEAM.createEset *ScreenBEAM.createEset: create eSet after mapping steps, including combining mapping info to your metadata and data normalization*

Description

ScreenBEAM.createEset: create eSet after mapping steps, including combining mapping info to your metadata and data normalization

Usage

```
ScreenBEAM.createEset(analysis.par, normalize.total = 1e+06, n.mismatch = 9)
```

Arguments

normalize.total,	integer
n.mismatch,	integer

ScreenBEAM.dir.create *ScreenBEAM.dir.create*

Description

ScreenBEAM.dir.create is used to help users create an organized working directory for the Functional Genomics Screening Projects. It creates a hierarchical working directory and returns a list contains this directory information.

Usage

```
ScreenBEAM.dir.create(
  project_main_dir = NULL,
  lib_name = NULL,
  DATE = TRUE,
  library_file = NULL,
  metadata_file = NULL,
  count_file = NULL,
  startFromFastq = TRUE
)
```

Arguments

project_main_dir	character, name or absolute path of the main working directory.
lib_name	character, name of the sh/sgRNA library. Each main working directory can contain multiple library sub-project folder.
DATE	logical, if TRUE, current date information will be added to the sub-project folder name. Default is TRUE.
library_file	character, path for the library file. Required when startFromFastq = TRUE.
metadata_file	character, path for the metadata csv file. (sampleID,group,replicate,sampleLabel,fastqFile)
count_file	character, path for the count tab file. (rnaID,geneID,Sample1,Sample2...). Required when startFromFastq = FALSE;
startFromFastq,	logical, if TRUE, the project start from Fastq files.

Details

This function needs users to define the main working directory and the library's name. It creates a main working directory with a subdirectory of the project(each library is a project). It also automatically creates 4 subfolders (fastq, metadata, library and output) within the project folder. fastq/, storing unzipped fastq files of each sample; metadata/, storing the experiment design, describe detailed information of each sample; library/, storing sh/sgRNA library csv file and its fasta file (for detail, please check demo). output/ folder contains 3 subfolder to store intermediate results generated by ScreenBEAM2 analysis. They are mapping/, QC/ and DR/.

Inside mapping/ there are 3 subfolders as well. mapping/Step1_unique_fast, mapping/Step2_blat and mapping/Step3_raw_count. mapping/Step1_unique_fast stores unique fasta read from each sample's fastq file. mapping/Step2_blat stores blat result when using unique fasta file and library fasta file. mapping/Step3_raw_count stores 3 raw count table created later in the pipeline. This

function also returns a list object (analysis.par in the demo) with directory information wrapped inside. This list is an essential for ScreenBEAM2 analysis, all the important intermediate data generated later will be wrapped inside.

QC/ is used to store quality control analysis of the mapping result. Including library check and mapping statistics. DR/ is used to store the Bayesian hierarchical meta analysis result.

If startFromFastq = FALSE, library_file is not required. Only metadata/, output/QC/ output/DR/ are created.

Value

Return a list object, containing all the paths of the created folder.

ScreenBEAM.gene.level	<i>GENE level meta-analysis of high-throughput Functional Genomics Screening analysis</i>
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Description

ScreenBEAM.gene.level takes a ".tsv" table file with samples raw count number, user needs to assign the control sample's name and case sample's name vector. It will do a gene-level meta-analysis of screening data. User also need to assign value to rna.size to the most common number of sh/sgRNAs targeting one gene. This function will remove bias caused by unbalanced number of sh/sgRNAs targeting the same gene.

Usage

```
ScreenBEAM.gene.level(
  input.file,
  control.samples,
  case.samples,
  control.groupname = "control",
  case.groupname = "treatment",
  gene.columnId = 2,
  data.type = c("microarray", "NGS"),
  do.normalization = TRUE,
  total = 1e+06,
  filterLowCount = FALSE,
  filterBy = "control",
  count.cutoff = 4,
  nitt = 15000,
  burnin = 5000,
  thin = 10,
  rna.size = 6,
  sample.rna.time = 100,
  method = "Bayesian",
  pooling = "partial",
  ...
)
```

Arguments

<code>input.file</code> ,	character, the file path of ".tsv" table with raw count number for each sample (column) and sh/sgRNA (row).
<code>control.samples</code> ,	vector of character, the control samples names, this information should match the column names of <code>input.file</code> . For example, <code>c(T0_A, T0_B)</code> .
<code>case.samples</code> ,	vector of character, the case samples names, this information should match the column names of <code>input.file</code> . For example, <code>c(T16_A, T16_B)</code> .
<code>control.groupname</code> ,	character, the name of your control samples' group name. For example, <code>T0</code> .
<code>case.groupname</code> ,	character, the name of your case samples' group name. For example, <code>T16</code> .
<code>gene.columnId</code> ,	integer, the number of the column which store the gene name. By default, 2.
<code>data.type</code> ,	character, can either be "microarray" or "NGS". Default is "microarray".
<code>do.normalization</code> ,	logical, if TRUE, a scaled normalization will be performed for each sample. To quantify this scale, user need to assign a value to <code>total</code> . Default is TRUE.
<code>total</code> ,	integer, need to be larger than the colSums of the raw count table. Default is <code>1e6</code> .
<code>filterLowCount</code> ,	logical, if TRUE, will remove rnas with low count, based on <code>filterBy</code> and <code>count.cutoff</code> . Default is TRUE.
<code>filterBy</code> ,	logical, if TRUE, will remove rnas with low count based on "control" samples or "case" samples. Default is "control".
<code>count.cutoff</code> ,	integer, the threshold of removing low count. Default is 4.
<code>nitt</code> ,	integer, the number of MCMC iterations. Default is 15000.
<code>burnin</code> ,	integer, burnin. Default is 5000.
<code>thin</code> ,	integer, thinning interval. Default is 10.
<code>rna.size</code> ,	integer, the most common number of sh/sgRNAs targetting the same gene. Default is 6.
<code>sample.rna.time</code> ,	integer, to remove unbalance number of sh/sgRNAs targetting the same gene. <code>rna.size</code> of sh/sgRNAs will be sampled, the sampling time.
<code>method</code> ,	character, estimation model. Either "Bayesian" or "MLE". Default is "Bayesian".
<code>pooling</code> ,	character, pooling method. Either "full" or "partial". Default is "partial".
<code>...</code>	parameter pass to function <code>DRAgeneLevel2</code>

Value

Return a data frame containing all the gene-level meta-analysis statistical values. Including `geneID`, `log2FC`, `z value`, `p value`, `FDR` and so on.

ScreenBEAM.mapping.QC *Create an HTML report to perform quality control of library, mapping status*

Description

ScreenBEAM.mapping.QC takes the master analysis.par list, which contains all the key data through ScreenBEAM meta-analysis. It will call Rmarkdown file to create an HTML report of library quality control (show the deleted duplicated/substring sh/sgRNA table). Percentage of mapping. Suggested rna.size, which is the most common number of sh/sgRNAs targeting the same gene. A barplot at the end shows the suggested number of mismatch for user to rescue the most reads.

Usage

```
ScreenBEAM.mapping.QC(
  analysis.par,
  QC.Rmd.path = system.file("Rmd", "mapping_QC_report.Rmd", package = "ScreenBEAM2")
)
```

Arguments

analysis.par, the master list, containing unique reads files path, blat result files path.
 QC.Rmd.path, character, the path where user put the Rmarkdown file of "mapping_QC_report".
 Default, it will call the one in ScreenBEAM2 package.

Value

It will create an Rmarkdown file and an HTML QC report in the folder of analysis.par\$out.dir.output.QC.

ScreenBEAM.Pairwise *GENE level meta-analysis of high-throughput Functional Genomics
Screening analysis for pairwise comparison*

Description

ScreenBEAM.Pairwise user needs to assign the control sample's name and case sample's name vector. It will do a gene-level/rna-level meta-analysis of screening data. User also need to assign value to rna.size to the most common number of sh/sgRNAs targeting one gene. This function will remove bias caused by unbalanced number of sh/sgRNAs targeting the same gene.

Usage

```
ScreenBEAM.Pairwise(
  analysis.par,
  choose_level = "gene",
  use_index,
  case_group,
  control_group,
```

```

gene.columnId = 2,
data.type = c("microarray", "NGS"),
do.normalization = TRUE,
total = 1e+06,
filterLowCount = TRUE,
filterBy = "control",
count.cutoff = 4,
nitt = 15000,
burnin = 5000,
thin = 10,
rna.size = 6,
sample.rna.time = 100,
method = "Bayesian",
pooling = "partial",
...
)

```

Arguments

analysis.par,	list
use_index,	integer, the line number for the 'analysis.par\$norm.path' to analyze.
case_group,	character, name for the case group
control_group,	character, name for the control group
gene.columnId,	integer, the number of the column which store the gene name. By default, 2.
data.type,	character, can either be "microarray" or "NGS". Default is "microarray".
do.normalization,	logical, if TRUE, a scaled normalization will be performed for each sample. To quantify this scale, user need to assign a value to total. Default is TRUE.
total,	integer, need to be larger than the colSums of the raw count table. Default is 1e6.
filterLowCount,	logical, if TRUE, will remove rnas with low count, based on filterBy and count.cutoff. Default is TRUE.
filterBy,	logical, if TRUE, will remove rnas with low count based on "control" samples or "case" samples. Default is "control".
count.cutoff,	integer, the threshold of removing low count. Default is 4.
nitt,	integer, the number of MCMC iterations. Default is 15000.
burnin,	integer, burnin. Default is 5000.
thin,	integer, thinning interval. Default is 10.
rna.size,	integer, the most common number of sh/sgRNAs targetting the same gene. Default is 6.
sample.rna.time,	integer, to remove unbalance number of sh/sgRNAs targetting the same gene. rna.size of sh/sgRNAs will be sampled, the sampling time.
method,	character, estimation model. Either "Bayesian" or "MLE". Default is "Bayesian".
pooling,	character, pooling method. Either "full" or "partial". Default is "partial".
...	parameter pass to function ScreenBEAM.gene.level

Value

Return analysis.par

ScreenBEAM.raw.count *Create raw count table for NGS data*

Description

ScreenBEAM.raw.count takes a list object, which contains all the paths of library information, unique fasta read path information and blat result path information. This function calculates basic statistics for the FASTQ raw reads, checks library quality, calculates mapping statistics and also collecting raw count numbe from blat result tables. This function can be run directly, or run step by step from its source code.

Usage

```
ScreenBEAM.raw.count(analysis.par)
```

Arguments

analysis.par list, the master list storing all the information along ScreenBEAM2 analysis.

Value

Return an updated list object, wrapping the raw count table information inside.

ScreenBEAM.rna.level *RNA level meta-analysis of high-throughput Functional Genomics Screening analysis*

Description

ScreenBEAM.rna.level takes a ".tsv" table file with samples raw count number, user needs to assign the control sample's name and case sample's name vector. It will do a rna-level meta-analysis of screening data.

Usage

```
ScreenBEAM.rna.level(
  input.file,
  control.samples,
  case.samples,
  control.groupname,
  case.groupname,
  gene.columnId = 2,
  filterLowCount = TRUE,
  filterBy = "control",
  count.cutoff = 16,
  do.normalization = TRUE,
```

```

total = 1e+06,
do.log2 = TRUE,
pooling = "full",
family = gaussian,
estimation.method = "Bayesian"
)

```

Arguments

`input.file`, character, the file path of ".tsv" table with raw count number for each sample (column) and sh/sgRNA (row).

`control.samples`, vector of character, the control samples names, this information should match the column names of `input.file`. For example, `c(T0_A, T0_B)`.

`case.samples`, vector of character, the case samples names, this information should match the column names of `input.file`. For example, `c(T16_A, T16_B)`.

`control.groupname`, character, the name of your control samples' group name. For example, T0.

`case.groupname`, character, the name of your case samples' group name. For example, T16.

`gene.columnId`, integer, the number of the column which store the gene name. By default, 2.

`filterLowCount`, logical, if TRUE, will remove rnas with low count, based on `filterBy` and `count.cutoff`. Default is TRUE.

`filterBy`, logical, if TRUE, will remove rnas with low count based on "control" samples or "case" samples. Default is "control".

`count.cutoff`, integer, the threshold of removing low count. Default is 16.

`do.normalization`, logical, if TRUE, a scaled normalization will be performed for each sample. To quantify this scale, user need to assign a value to `total`. Default is TRUE.

`total`, integer, need to be larger than the colSums of the raw count table. Default is 1e6.

`do.log2`, logical, if TRUE, the expression data will be log2 transformed.

`pooling`, character, pooling method. Either "full" or "partial". Default is "full".

`family`, function, default if gaussian.

`estimation.method`, character, estimation model. Either "Bayesian" or "MLE". Default is "Bayesian".

Value

Return a data frame containing all the rna-level meta-analysis statistical values. Including sh/sgRNA ID, log2FC, p.value and so on.

`ScreenBEAM.trim.helper`*Create a pdf plot to visualize the trim location.*

Description

`ScreenBEAM.trim.helper` is a helper function for users who wants to know the most common trim place in their FASTQ reads. This function requires the linux command `seqtk`. Please make sure you have it before calling this function.

Usage

```
ScreenBEAM.trim.helper(fastq.path, sample_num = 10000, pdf.file)
```

Arguments

<code>fastq.path</code> ,	character, the file path of one FASTQ reads file. Unzipped.
<code>sample_num</code> ,	integer, the random sample number of FASTQ reads when creating the barplot. Default is 10000.
<code>pdf.file</code> ,	character, the file path to store the output pdf plot.

Value

It will create a PDF barplot file showing A,T,C,G and N's percentage at each base position. User can pick the trim location based on the plot.

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